

Memristors And Their Application In NEUROMORPHIC COMPUTING

Venturing into memristor technology brings us nearer to brain-like AI. This leap could reshape how we think about computing.



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Memristors, a portmanteau of ‘memory’ and ‘resistor,’ are a groundbreaking class of passive two-terminal circuit elements. First theorised by Leon Chua in 1971 and physically realised by HP Labs in 2008, memristors retain a state of resistance based on the history of voltage and current passed through them. This unique property makes them similar to synapses in the human

brain, enabling them to play a crucial role in neuromorphic computing.

Fundamental principles of memristors

At their core, memristors operate on the principle that their resistance can change when an electric current is applied. This resistance remains until another current is applied to change it

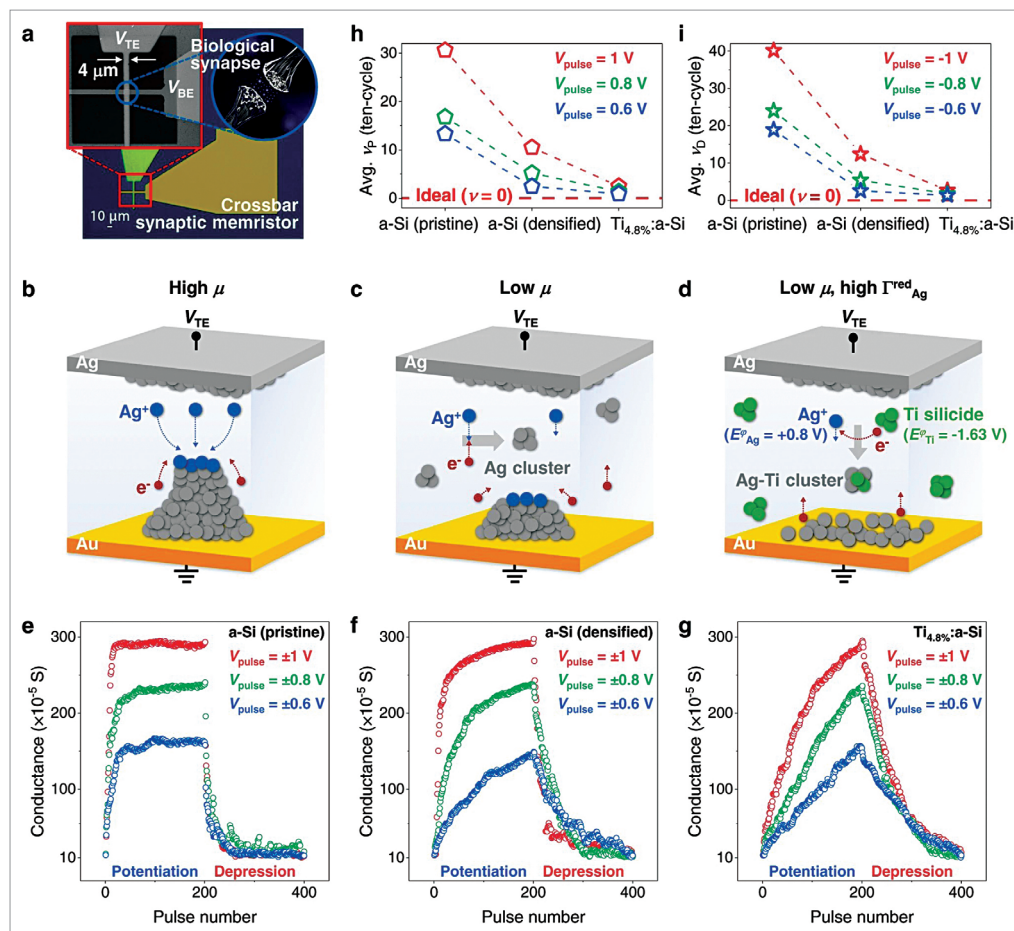


Fig. 1: The effect of cluster-type switching (Credit: <https://www.nature.com>)